

Implementation of Untethered Biped Robots Utilizing Serial-Parallel Hybrid Leg Mechanisms

Kenta Hirashima¹, Noboru Myers¹, Daniel Campos Zamora², Kevin G. Gim¹, and Joohyung Kim¹

Abstract—In this paper, we present a new, standalone bipedal robot system that features Hybrid Leg linkages and a 2-DOF neck to incorporate a camera. An online walking pattern generator using ZMP preview control enables remote robot operation based on user commands. The foot trajectory is modified by IMU sensor feedback for stabilization. Taking advantage of the low structural inertia of Hybrid Leg, the effectiveness of bipedal locomotion with pattern generation based on the cart-table model is tested. Given the bio-inspired nature of the Hybrid Leg structure, the gait pattern is chosen to emulate that of humans. We demonstrate bipedal locomotion on an approximately flat table while controlling it remotely. Simultaneous control of two Hybrid Leg bipedal robots is also shown to highlight the performance of the system.

I. INTRODUCTION

A humanoid robot's unique ability to walk on its two legs depends largely on good walking pattern generation and stabilization control. One widely used criterion for generation of a stable walking pattern is Zero Moment Point (ZMP). It was first introduced by Vukobratović and Stepanenko, and its possible application to bipedal gait synthesis was discussed in [1]. The concept of ZMP had been utilized on various humanoid robots. WL-12RV from Waseda University used pre-planned ZMP trajectory with trunk motion for ZMP compensation [2]. Honda's humanoid robot P2 also generated walking pattern based on desired ZMP trajectory [3].

While these approaches rely on accuracy of the robot model, some take a different approach with a reduced order model. Introduced by Kajita et al. [4], the 3D Linear Inverted Pendulum Mode (3D-LIPM) based walking pattern generation aims to plan a Center of Mass (CoM) trajectory that considers a simplified humanoid model, an inverted pendulum. Although this approach does not explicitly use the concept of ZMP in the pattern generation, the modeling approach captures the dynamics in such a way that the underlying ZMP motion influences the resultant CoM motion. Kajita et al. introduced another method of humanoid gait generation which utilizes the cart-table model [5]. This modeling approach takes a reversed view on the input and output relationship of ZMP and CoM, enabling pattern generation based on the desired ZMP, from which the necessary CoM movement can be determined. This can be done off-line with short computation time in discrete-time domain [6], but preview

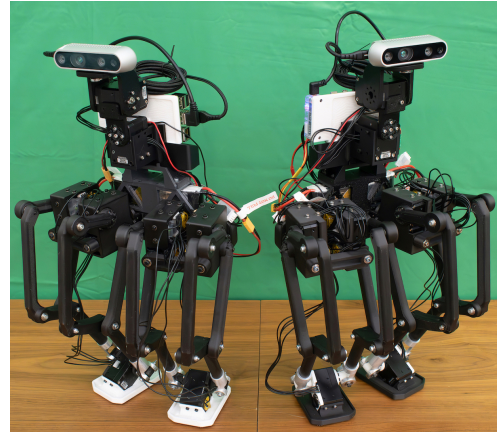


Fig. 1. Two prototypes of Hybrid Leg bipedal robot

control allows for on-line walking pattern generation. More precisely, with an assumption that future ZMP reference is provided, an optimal preview controller [7] suitably generates a CoM trajectory by framing the problem as a ZMP tracking control.

This gait synthesis was successfully implemented on the full-sized humanoid HRP-2 [8]. Some smaller scale robot systems that incorporate ZMP preview control were developed as well. [9] simulated ZMP preview control-based walking pattern generation on a robot called mini hubo, but it lacked hardware experimentation. Preview control of ZMP has been implemented on the small scale humanoid Nao as well [10], but it has a vastly different leg structure compared to our humanoid which uses a serial-parallel linkage mechanism called *Hybrid Leg* [11]–[13].

A Hybrid Leg consists of a pair of 3 Degrees of Freedom (DOF) chains connected in parallel at the ankle. Each chain is a 5-bar linkage mechanism with two servo motors, which realize hip pitch and knee pitch motion. Both servo motors are located close to the hip in order to achieve a low leg inertia. At the connection of the two chains, another servo is located for ankle pitch motion.

In our previous work, an analytical solution to the kinematics of the Hybrid Leg was determined and performance of the leg design was evaluated through workspace and inverse dynamics analysis. Furthermore, the lower body of a bipedal robot was assembled using two Hybrid Legs, and a bipedal walking test was performed with manually generated foot position commands [12]. However, the robot was tethered and joint commands as well as power were provided through a wire.

¹Kenta Hirashima, Noboru Myers, Kevin G. Gim, and Joohyung Kim are with the KIMLAB (Kinetic Intelligent Machine Lab), University of Illinois Urbana-Champaign, IL 61801, USA. ²Daniel Campos Zamora is with University of Washington, Seattle, WA 98195, USA. Author contact information is {kentah2, joohyung}@illinois.edu

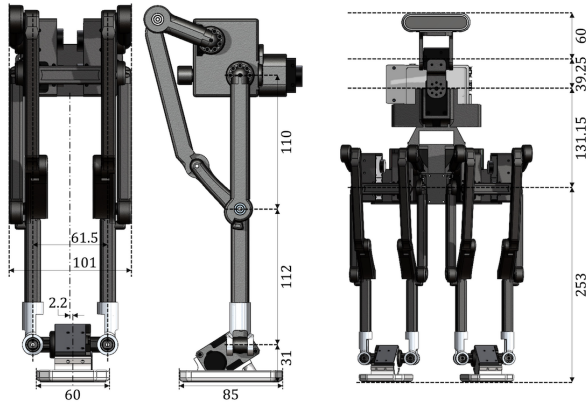


Fig. 2. Dimensions of a proposed leg and overall system (dimensions in mm)

In this paper, we fabricate two new prototypes of the bipedal robots using Hybrid Legs with linkages that are 3D printed with carbon fiber infused filament. Each robot is made stand-alone by using LiPo batteries as a power source and a single board computer (SBC) Raspberry Pi 4 model B with 8 GB of memory. Additionally, a 2-DOF head is attached to the body to incorporate an Intel®RealSense camera D435. For controlling the robot, bipedal gait generation based on preview control of ZMP is implemented, and the generated gait is modified based on inertial measurement unit (IMU) sensor feedback for stabilization control.

The rest of this paper is organized as follows. In Section II, we detail the modifications and improvements from the previous Hybrid Leg design and the implementation of the full bipedal system hardware. In Section III, we present the trajectory generation and walking controls using the ZMP criterion and the cart-table model. In Section IV, we implement IMU based stabilization control and demonstrate bipedal walking. Finally, in Section V we discuss conclusions and future works.

II. HARDWARE IMPLEMENTATION

A. Modified Hybrid Leg Design

The early prototype of Hybrid Leg was constructed from fully 3D printed links. This proved to be less favorable due to high friction at joints and ductility of slender linkages. Hence, in the previous bipedal platform, carbon fiber tubes and 3D printed bearing carriage tips were used. The new prototype of Hybrid Leg consists of links that are 3D printed with carbon fiber reinforced Nylon filament, which helps reduce the ductility of each link and contributes to overall rigidity of the leg. The ankle link is machined aluminum and the foot piece for one of the two humanoid prototypes is also 3D printed with regular, white Polylactic Acid (PLA) filament as well.

To further eliminate the structural compliance, another link is added between servo motor housings. This constrains the roll axes of the left and right linkages. The lower leg link length and foot size have also been modified. The width of the leg is reduced and the length is increased to improve

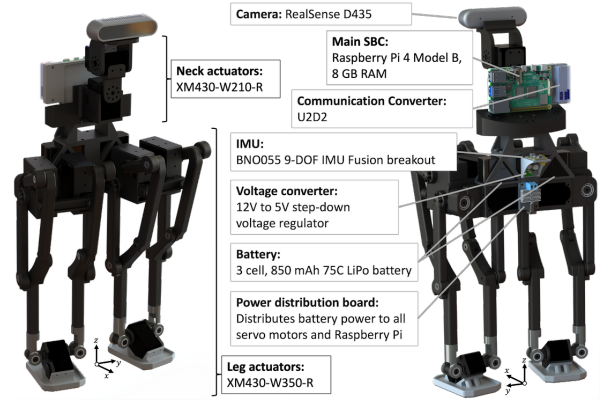


Fig. 3. Hardware specification of the proposed system

the workspace of the leg. Similarly, the center of the two linkages and the center of the foot are offset by 2.2 mm to minimize collisions of feet during walking. Since the modeling of bipedal robot utilized for this study assumes a non-slip condition at foot contact, a non-slip tape for furniture is applied at the sole. Finally, each leg is actuated by six servo actuators (DYNAMIXEL XM430-W350-R).

B. Hybrid Leg Bipedal System

Other than design changes of the leg, the new bipedal system has several features and components added to the previous prototype. The body now consists of a truss structure sitting on top of the hip structure that connects two Hybrid Legs. As in the previous prototype, the legs are oriented with 10° toe-out angle. This design was inspired by an observation that an average male adult has a roughly 11° toe-out angle in a normal walking gait [14]. The truss structure houses two 850 mAh 3-cell LiPo batteries with 75 C rating. The two batteries are connected in parallel and allow for 30 minutes to an hour of run time.

The same batteries also supply power to the SBC, through which the IMU sensor and camera are powered. The system on chip (SoC) is a Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit at 1.8 GHz. The operating system is Ubuntu Linux version 20.04.5 LTS and uses ROS noetic along with DynamixelSDK as the middleware. All software implementations are done using C++. The IMU sensor is Adafruit BNO055 9-DOF absolute orientation fusion breakout. It is capable of fusing the sensor data from accelerometer, gyroscope, and magnetometer to output the absolute orientation. The IMU communicates with the SBC via I2C protocol and sends quaternion orientation that is later converted to Euler angles for feedback calculation. The camera is capable of capturing images at 15 frames per second (FPS) while walking.

The head is attached to the top of the truss structure and houses the SBC, USB2DYNAMIXEL connector (U2D2), and a 2-DOF neck on which the camera is mounted. The neck uses two servo motors (DYNAMIXEL XM430-W210-R) and allows the camera to rotate in the roll and pitch directions relative to the hip frame.

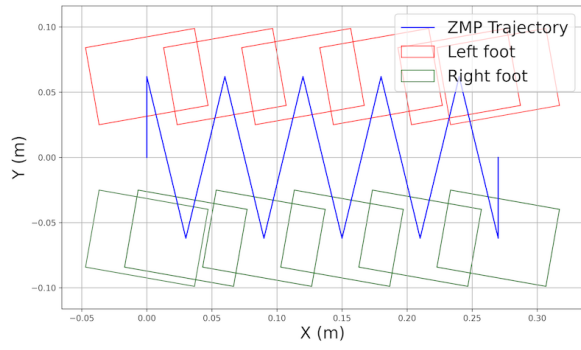


Fig. 4. Reference ZMP trajectory and desired foot placement for 30 mm step length and 62 mm step width

III. SOFTWARE IMPLEMENTATION

A. Reference ZMP Generation

ZMP criterion is a metric to determine the stability of a walking robot. Hence, one common method to realize bipedal walking is based on planning of a desired ZMP trajectory and tracking it by appropriate placement of the supporting foot. In this paper, it is assumed that frontal and sagittal motions can be separated, thus, the reference ZMP trajectory is generated separately for the x and y directions. The implemented algorithm generates a reference ZMP trajectory based on a given step time t_{step} , double support phase (DSP) duration t_{dbl} , step width, and step length. Here, the step time refers to half the duration of the gait cycle.

The reference ZMP trajectory and the corresponding desired foot positions are designed so that the ZMP stays within the convex hull of the support polygon. To give the largest margin of stability, the ZMP should stay at the center of the supporting foot during single support phase (SSP). During DSP, the ZMP transitions from one foot to another. While the robot is stationary, the ZMP is located in the middle of the centers of the left and right foot. Keyboard input signals the ZMP reference generator to either start or stop the reference generation. An internal algorithm constrains the timing for gait changes even if the keyboard input is not synchronized.

Figure 4 shows an example of such a reference ZMP trajectory generated. The step length for this trajectory is 30 mm and the step width, the displacement of foot's center from center of hip, is 62 mm. As illustrated, the toe-out angle of both feet are maintained at 10° . Moreover, the step time is 0.3 seconds and the DSP duration is 0.06 seconds, which makes the stance phase to swing phase ratio 3:2. This coincides with the normal human gait cycle consisting of a 60% stance phase, 40% swing phase, and DSP that each last about 10% of gait cycle [15].

B. ZMP-based walking pattern generation

Representing the complicated physical system as a reduced order model is a viable tactic for simplifying its dynamics and control strategy. The 3D-LIPM [16] is one such model that represents a humanoid as a simple inverted pendulum.

Assuming the mass is concentrated at the robot CoM and the CoM height stays constant at z_c , the equation of motion of LIPM in the sagittal plane is given by,

$$\ddot{x} = \frac{g}{z_c}(x - p_x), \quad (1)$$

where p_x denotes the position of the ZMP, x denotes the position of the CoM, and g is the gravitational acceleration. The same exact equation holds for motion in the frontal plane. This model implies that change in the CoM is caused by the ZMP motion. Consequently, pattern generation based on 3D-LIPM first determines the target CoM trajectory and plans the ZMP trajectory that realizes the desired CoM trajectory [4]. However, this method generates a stable gait at a cost of modifying foot step locations, which makes the approach unsuitable for a gait with specific foot placements [5]. The cart-table model overcomes this issue by capturing the dynamics in a reversed manner. Namely, a change in the ZMP is caused by CoM movement. Assuming the mass-less table and the constant CoM height at z_c , the equation of motion for the cart-table model is given by,

$$p_x = x - \frac{z_c}{g}\ddot{x}. \quad (2)$$

The walking pattern generation then becomes a task of determining desired ZMP motion and solving for a CoM trajectory that realizes that desired ZMP motion [17].

C. ZMP Preview Controller

Solving for the CoM trajectory is done by framing it as a tracking control of ZMP with cart-table model. The input to the system is defined as the jerk of the cart. Hence,

$$u = \ddot{\ddot{x}}. \quad (3)$$

The corresponding state space representation in this case is

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u, \\ p &= \begin{bmatrix} 1 & 0 & \frac{-z_c}{g} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix}. \end{aligned} \quad (4)$$

To enable digital implementation of the control scheme, Eq. 4 is discretized by the control period Δt as

$$\begin{cases} \mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{b}u_k, \\ p_k = \mathbf{c}\mathbf{x}_k. \end{cases} \quad (5)$$

The state space matrices are defined as follows:

$$\begin{aligned}
\mathbf{x}_k &\equiv [x(k\Delta t) \quad \dot{x}(k\Delta t) \quad \ddot{x}(k\Delta t)]^T, \\
u_k &\equiv u(k\Delta t), \\
p_k &\equiv p(k\Delta t), \\
\mathbf{A} &\equiv \begin{bmatrix} 1 & \Delta t & \Delta t^2/2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}, \\
\mathbf{b} &\equiv \begin{bmatrix} \Delta t^3/6 \\ \Delta t^2/2 \\ \Delta t \end{bmatrix}, \\
\mathbf{c} &\equiv [1 \quad 0 \quad -z_c/g].
\end{aligned}$$

By denoting the ZMP reference as $p^{ref}(k)$, the performance index J is given by,

$$J = \sum_{i=k}^{\infty} Q_e (p_i - p_i^{ref})^2 + \Delta \mathbf{x}_i^T \mathbf{Q}_x \Delta \mathbf{x}_i + R \Delta u_i^2, \quad (6)$$

where $\Delta \mathbf{x}_k \equiv \mathbf{x}_k - \mathbf{x}_{k-1}$ and $\Delta u_k \equiv u_k - u_{k-1}$ are the incremental state vector and incremental input, respectively. Also, the weight for tracking error Q_e and the weight for input R are greater than 0. The weight for states Q_x is a 3×3 symmetric non-negative definite matrix. The weights selected for this study are $Q_e = 100.0$, $Q_x = 0$, and $R = 1.0 \times 10^{-5}$. As shown in [7], the optimal controller that minimizes Eq. 6 with previewable reference of up to N step future is given by

$$u(k) = -G_i \sum_{i=0}^k (p_k - p_k^{ref}) - G_x \mathbf{x}_k - \sum_{j=1}^N G_{p,j} p_{k+j}^{ref}. \quad (7)$$

The optimal gains G_i , G_x , and G_p are calculated by solving Discrete-time Algebraic Riccati Equation (DARE). While a programming platform like MATLAB® offers a convenient function for calculating solutions to DARE, it could an exhausting process for development in C++. The structure-preserving doubling algorithm (SDA) introduced in [18] is a DARE solver algorithm that has fast convergence. With the use of Eigen library [19], it becomes suitable for implementation in C++. Hence, the algorithm is adopted for software implementation in this study. For a set of fixed walking parameters, the DARE can be computed offline. While the parameters are indeed fixed for this research, online adjustment of walking parameters may be required in the future. Thus, computation of DARE is done online as well.

For determining appropriate preview time, the robot's CoM position is constrained to be at $z_c = 0.2309$ m. By design, this aligns with the z-position of CoM of the robot at the initial pose. The control loop period is $\Delta t = 8$ ms. The preview gain G_p calculated with the given parameters is small enough for a preview time beyond 1.4 s. Thus, the preview time $t_{preview}$ is chosen to be 1.4 s for this study. Reference ZMP and CoM trajectory for motion in sagittal plane and frontal plane are shown in Fig. 5.

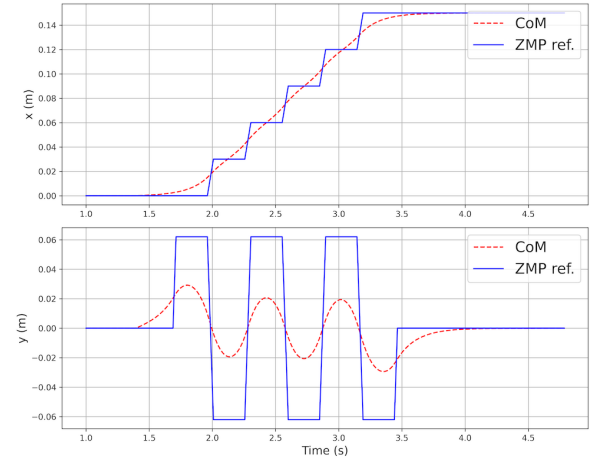


Fig. 5. Reference ZMP and calculated CoM trajectory for sagittal and frontal motion that corresponds to 5 steps forward

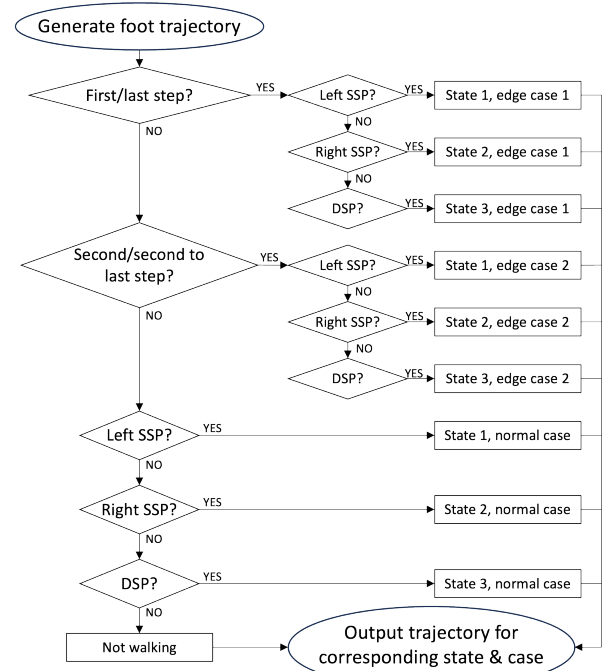


Fig. 6. Algorithm flowchart for foot trajectory generation

D. Foot trajectory generation

The goal of foot trajectory generation is to achieve appropriate placement of the swing foot while minimizing landing impact. The x and y offset of CoM from the center of hip is calculated at the initial pose and is assumed to stay constant, allowing the the CoM trajectory to be approximated as the hip trajectory. The offset is applied when passing foot position trajectory to the inverse kinematics.

For generating a foot trajectory that is capable of transitioning between different phases, a finite number of walking states are defined. The states are as follows.

- 1) Left SSP (right swing phase)

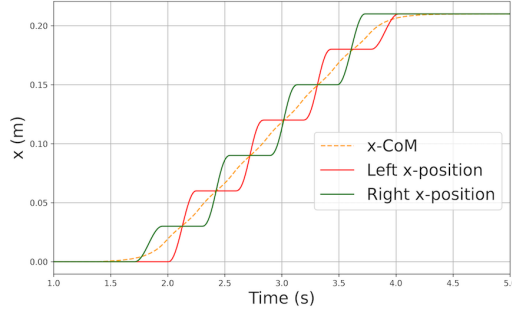


Fig. 7. x-position of feet and CoM

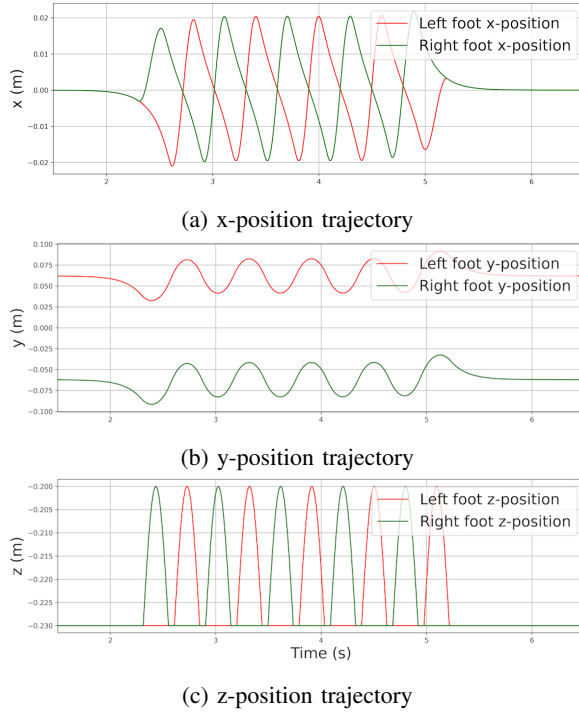


Fig. 8. Plots showing x, y, and z foot position trajectories for 9-step forward walking, calculated with respect to hip frame

- 2) Right SSP (left swing phase)
- 3) Left or right DSP

Furthermore, two edge cases are defined for 1) the first step or last step and 2) second step or second to last step. The summary of this finite state machine is shown in Fig. 6.

Edge cases are defined because the x-direction foot trajectory is calculated by combining a sinusoidal curve and CoM trajectory, which does not settle to simple harmonic motion until after the first two steps. Similarly, when stopping, the simple harmonic oscillation of the CoM is useful for foot trajectory calculation only up to the last two steps. This is apparent in Fig. 7 which shows a plot of x-position of the left foot, right foot, and CoM all with respect to the coordinate frame fixed to the starting position. For the edge cases, CoM positions are not considered in foot trajectory generation, and a simple cosine curve is used. The calculated x-position

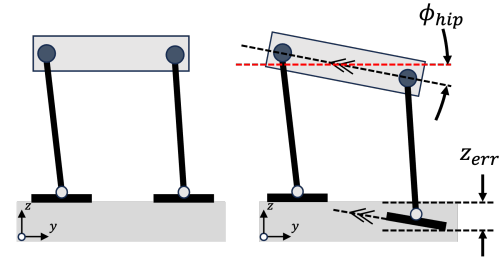


Fig. 9. Illustration of hip roll offset θ_{hip} and consequent error in foot z-position z_{err}

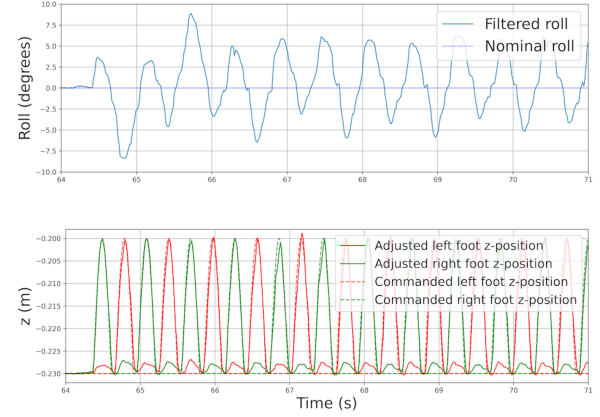


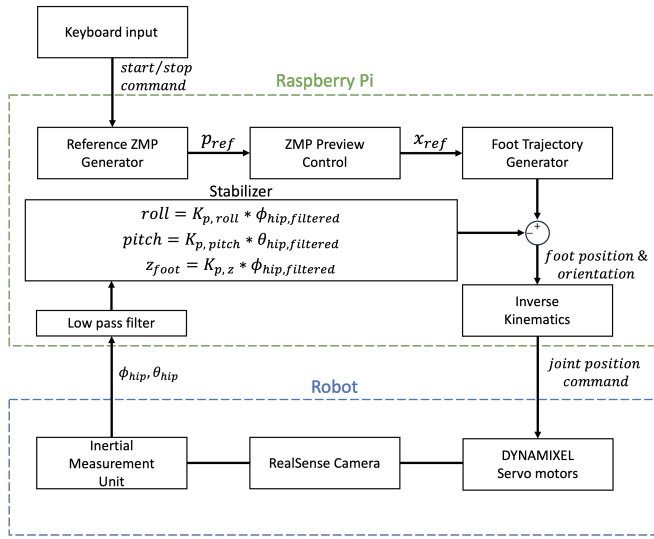
Fig. 10. Plots of filtered roll (top) and the reference vs. adjusted z-position trajectory (bottom) during a forward walking

trajectory is transformed to be expressed with respect to the hip frame.

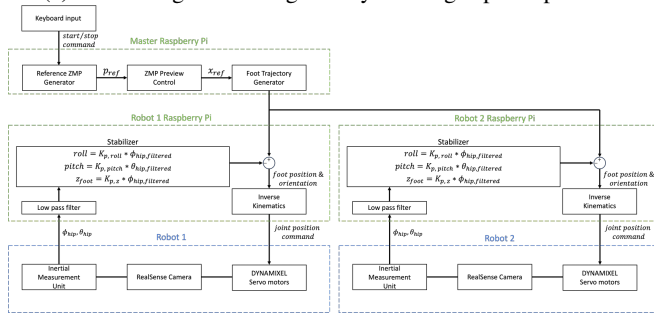
Z-position trajectory is generated using the same finite state machine, but without any edge cases. It is a sine curve with an amplitude equal to the amount by which to raise the foot and a period of twice the duration of the SSP. The y-position trajectory is calculated simply by inverting the CoM's y-position trajectory and giving an offset equal to the step width. The foot position trajectories that would correspond to the reference ZMP in Fig. 4 are shown in Fig. 8. A z-position of a foot of -0.23m indicates that the foot should be in contact with the ground.

IV. BIPEDAL LOCOMOTION

In addition to the implementation of walking pattern generator described in the previous section, a simple stabilization control based on IMU feedback is added for online modification of the given foot position trajectory. [20] implemented an upright pose controller for walking on uneven terrain. Although the scope of this study is walking on even terrain, a similar controller is implemented for stabilization control. The foot roll and pitch are commanded to stay at 0 degrees with respect to the pelvis, which is assumed to stay parallel to the ground at all times. However, such an assumption does not always hold due to a lack of servo power or suboptimal tracking performance of the hip roll joint. This can cause



(a) Block diagram of singular Hybrid Leg bipedal platform

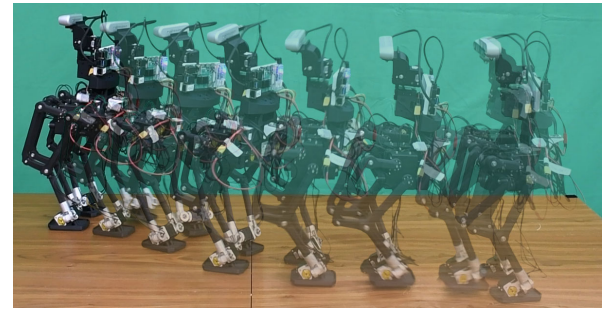


(b) Block diagram for system of two Hybrid Leg bipedal platforms

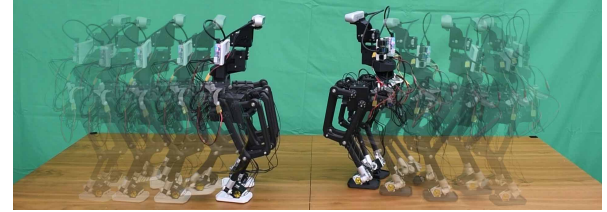
Fig. 11. Block diagrams for a system of (a) singular robot and (b) two robots

poor foot contact with the ground, making the robot unstable. Fig. 9 illustrates such a situation. Furthermore, because the z-position of the foot is defined with respect to the hip center, the hip roll offset also necessitates a modification of the z-position command to avoid over-extending the leg. To minimize the error of z-position, z_{err} , a z-height compensator is implemented to modify the z-position command.

The z-height compensator takes data from the IMU to estimate the current hip orientation. Since the raw orientation data is noisy, it is passed through a second-order butterworth lowpass filter with a 20 Hz cutoff frequency. The filtered IMU data is used to adjust the Z-height of the foot depending on the hip orientation. Furthermore, a foot orientation controller similar to the one proposed in [20] is also implemented. Recording the initial foot orientation at standing pose as being parallel to the hip and thus the ground, the filtered IMU data is used to adjust the foot orientation so that it stays parallel to the ground. These stabilization controllers aim to minimize the landing impact and keep the robot from becoming unbalanced. The block diagram for the implemented walking pattern generator and stabilization controller for a singular robot is shown in Fig. 11a. The proportional gains $K_{p,roll}$, $K_{p,pitch}$, and $K_{p,z}$ are



(a) Overlaid snapshots of steering motion



(b) Overlaid snapshots of two robots experiment

Fig. 12. Overlaid snapshots of various experiments

heuristically determined. In the case of a system of two robots, an additional SBC is prepared and dedicated for pattern generation as a master. The master SBC publishes a foot trajectory topic to which the two robots subscribe. The corresponding system diagram is shown in Fig. 11b.

To change the walking direction of the robot, keyboard input is used for changing the yaw orientation of the supporting foot. This simply rotates the supporting foot in the opposite direction to the desired steering direction, hence rotating the robot to the correct direction. The yaw rotation of a foot occurs during the SSP and the yaw angle changes linearly by a given yaw rotation angle. The foot is then rotated back to the nominal yaw orientation during the swing phase.

To validate the effectiveness of the implemented walking pattern generation and stabilizing controller, a bipedal walking experiment is conducted on an approximately flat dinner table. The parameters for pattern generation are the same as shown in Fig. 4: $t_{step} = 0.3$ s, $t_{dbl} = 0.06$ s, $t_{preview} = 1.4$ s, and maximum z-displacement of the swing foot is 30 mm off the ground. The robot was remotely controlled by keyboard input and walked for 38 steps on the table without falling down. Experiments involving turning motion are also performed. The robot was able to turn in-place with a slight drift of position. The yaw rotation angle was set to 20° for in-place turning experiment. Additionally, the robot successfully steered while walking forward based on keyboard input. The yaw rotation angle was set to 10° for steering experiment. The overlaid snapshots for this experiment are shown in Fig. 12a. The walking performance worsened as the batteries drained. Frequent falls were seen when the voltage dropped to around the LiPo cell's nominal voltage, which typically occurred 20 to 30 minutes after the start of an experiment.

Another experiment is conducted with two Hybrid Leg robots using the protocol shown in Fig. 11b. The overlaid

snapshots for this experiment is shown in Fig. 12b. Due to a communication lag and subsequent jerky foot position tracking, one of the robots drifts to its right. Nevertheless, both robots walked towards each other and stopped after taking 14 steps. Please refer to the supplemental video for the robot's walking experiments.

V. CONCLUSIONS AND FUTURE WORK

In this paper, the design of Hybrid Leg is modified and the new bipedal platform is fabricated. The platform is made stand-alone by using LiPo battery power and a Raspberry Pi as the main computer running Ubuntu Linux 20.04. Using ROS, walking pattern generation based on preview control of ZMP is developed. Furthermore, the stabilization controller modifies the calculated foot position trajectories by taking the robot's orientation data from an IMU sensor attached to the pelvis. The controller along with ZMP-based pattern generation is shown to successfully achieve forward walking on approximately flat ground. Furthermore, a system of two robots was implemented and they were simultaneously controlled to walk towards each other.

For our future work, we plan on improving the controller by using ZMP feedback and the 3D-LIPM model to make appropriate adjustments to the CoM trajectory online. We will also explore a reinforced learning approach for improvement of walking performance. The gait cycle equivalent to that of a typical human is used for the bipedal pattern generation in this paper, but a learning based approach may allow for more flexible tuning of the parameters specialized for the Hybrid Leg.

On the hardware side, the structural rigidity of the parallel link mechanism should be enhanced. Play in the linkage mechanism likely created by the passive hip joint on Hybrid Leg negatively affected the foot position tracking under harsh loading conditions. This can be overcome with some modifications to the mechanism while not affecting the workspace and not adding significant additional weight.

REFERENCES

- [1] M. Vukobratović and J. Stepanenko, "On the stability of anthropomorphic systems," *Mathematical Biosciences*, vol. 15, no. 1, pp. 1–37, 1972. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0025556472900612>
- [2] J.-I. Yamaguchi, A. Takanishi, and I. Kato, "Development of a biped walking robot compensating for three-axis moment by trunk motion," in *Proceedings of 1993 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS '93)*, vol. 1, 1993, pp. 561–566 vol.1.
- [3] K. Hirai, M. Hirose, Y. Haikawa, and T. Takenaka, "The development of honda humanoid robot," in *Proceedings. 1998 IEEE International Conference on Robotics and Automation*, vol. 2, 1998, pp. 1321–1326 vol.2.
- [4] S. Kajita, O. Matsumoto, and M. Saigo, "Real-time 3d walking pattern generation for a biped robot with telescopic legs," in *Proceedings 2001 ICRA. IEEE International Conference on Robotics and Automation*, vol. 3, 2001, pp. 2299–2306 vol.3.
- [5] S. Kajita, F. Kanehiro, K. Kaneko, K. Fujiwara, K. Harada, K. Yokoi, and H. Hirukawa, "Biped walking pattern generation by using preview control of zero-moment point," in *2003 IEEE International Conference on Robotics and Automation*, vol. 2, 2003, pp. 1620–1626 vol.2.
- [6] S. Kagami, K. Nishiwaki, T. Kitagawa, T. Sugihara, M. Inaba, and H. Inoue, "A fast generation method of a dynamically stable humanoid robot trajectory with enhanced zmp constraint," in *Proc. of IEEE International Conference on Humanoid Robotics (Humanoid2000)*, 2000.
- [7] T. Katayama, T. OHKI, T. INOUE, and T. KATO, "Design of an optimal controller for a discrete-time system subject to previewable demand," *International Journal of Control - INT J CONTR*, vol. 41, pp. 677–699, 03 1985.
- [8] K. Nishiwaki and S. Kagami, "High frequency walking pattern generation based on preview control of zmp," in *Proceedings 2006 IEEE International Conference on Robotics and Automation, 2006. ICRA 2006.*, 2006, pp. 2667–2672.
- [9] Y. Jun, R. Ellenberg, and P. Oh, "Realization of miniature humanoid for obstacle avoidance with real-time zmp preview control used for full-sized humanoid," in *2010 10th IEEE-RAS International Conference on Humanoid Robots*, 2010, pp. 46–51.
- [10] J. Strom, G. Slavov, and E. Chown, "Omnidirectional walking using zmp and preview control for the nao humanoid robot," in *RoboCup 2009: Robot Soccer World Cup XIII*, J. Baltes, M. G. Lagoudakis, T. Naruse, and S. S. Ghidary, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010, pp. 378–389.
- [11] K. G. Gim, J. Kim, and K. Yamane, "Design of a serial-parallel hybrid leg for a humanoid robot," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 6076–6081.
- [12] —, "Design and fabrication of a bipedal robot using serial-parallel hybrid leg mechanism," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 5095–5100.
- [13] K. G. Gim and J. Kim, "Implementation of a large-scale biped robot using serial-parallel hybrid leg mechanism," in *2022 19th International Conference on Ubiquitous Robots (UR)*, 2022, pp. 115–121.
- [14] H. Koblauch, T. Heilskov-Hansen, T. Alkjær, E. B. Simonsen, and M. Henriksen, "The effect of foot progression angle on knee joint compression force during walking," *Journal of Applied Biomechanics*, vol. 29, no. 3, pp. 329 – 335, 2013.
- [15] M. W. Whittle, "Chapter 2 - normal gait," in *Gait Analysis (Fourth Edition)*, fourth edition ed., M. W. Whittle, Ed. Edinburgh: Butterworth-Heinemann, 2007, pp. 47–100.
- [16] S. Kajita, F. Kanehiro, K. Kaneko, K. Yokoi, and H. Hirukawa, "The 3d linear inverted pendulum mode: a simple modeling for a biped walking pattern generation," in *Proceedings 2001 IEEE/RSJ International Conference on Intelligent Robots and Systems. Expanding the Societal Role of Robotics in the the Next Millennium*, vol. 1, 2001, pp. 239–246 vol.1.
- [17] S. Kajita, H. Hirukawa, K. Harada, and K. Yokoi, *Biped Walking*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2014, pp. 105–158. [Online]. Available: https://doi.org/10.1007/978-3-642-54536-8_4
- [18] W.-W. L. E. K.-W. Chu, H.-Y. Fan and C.-S. Wang, "Structure-preserving algorithms for periodic discrete-time algebraic riccati equations," *International Journal of Control*, vol. 77, no. 8, pp. 767–788, 2004. [Online]. Available: <https://doi.org/10.1080/00207170410001714988>
- [19] G. Guennebaud, B. Jacob *et al.*, "Eigen v3," <http://eigen.tuxfamily.org>, 2010.
- [20] J.-Y. Kim, I.-W. Park, and J.-H. Oh, "Walking control algorithm of biped humanoid robot on uneven and inclined floor," *Journal of Intelligent and Robotic Systems*, vol. 48, no. 4, pp. 457–484, Apr. 2007.